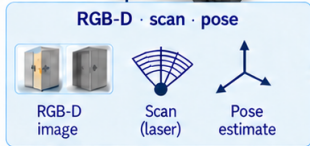


(a) Target platform

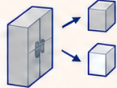


(b) CPU virtual validation

Global observation planning

OV-SDF

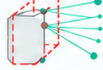
Semantic evidence



Category prior

IGCR

Geometric feedback



-- Predicted (template)

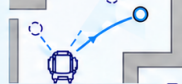
— Measured (scan)

Residual correction



STGHP

Budgeted view planning



○ Candidate view

● Selected next pose

→ Planned route



Next pose

Local execution

RPN-UQ

Action & return guard



Execute

Move / turn



Next observation

Acquired data

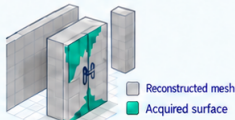
(c) Measured reconstruction

2D occupancy



From scan + pose

CPU TSDF



From depth + pose (ZED)

Saved map + mesh

Offline evaluation



Reference geometry



Coverage × surface F_1